

On the Accuracy–Complexity Tradeoff of Fuzzy Broad Learning System

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前件只用了一个模糊子系统，后件就是简单的增量学习

Abstract—The fuzzy broad learning system (FBLS) is a recently proposed neuro-fuzzy model that shares the similar structure of a broad learning system (BLS). It shows high accuracy in both classification and regression tasks and inherits the fast computational nature of a BLS. However, the ensemble of several fuzzy subsystems in an FBLS decreases the possibility of understanding the fuzzy model since the fuzzy rules from different fuzzy systems are difficult to combine together while keeping the consistency. To balance the model accuracy and complexity, a synthetically simplified FBLS with better interpretability, named compact FBLS (CFBLS), is developed in this article, which can generate much fewer and more explainable fuzzy rules for understanding. In such a way, only one traditional Takagi–Sugeno–Kang fuzzy system is employed in the feature layer of a CFBLS, and the input universe of discourse is equally partitioned to obtain the fuzzy sets with proper linguistic labels accordingly. The random feature selection matrix and rule combination matrix are employed to reduce the total number of fuzzy rules and to avoid the “curse of dimensionality.” The enhancement layer is kept in the CFBLS that helps to add a nonlinear transformation of input features to the traditional first-order polynomial used in the consequent part of a fuzzy rule. The pseudoinverse is also used to determine the parameters of CFBLS guaranteeing its fast computational nature. The experiments on the popular UCI and KEEL datasets indicate that the CFBLS can generate a smaller set of comprehensible fuzzy rules and achieve much higher accuracy than some state-of-the-art neuro-fuzzy models. Moreover, the advantage of CFBLS is also verified in a real-world application.

Index Terms—Accuracy–complexity, classification, fuzzy broad learning system (FBLS), Takagi–Sugeno–Kang (TSK) fuzzy system.

I. INTRODUCTION

NEURO-FUZZY models which are also called fuzzy neural networks have always been an active field that attracts the attention of researchers since the representative framework adaptive-network-based fuzzy inference system [1] was proposed. The neuro-fuzzy model is believed to inherit the merits of fuzzy systems and neural networks: The interpretable IF–THEN fuzzy rules and trainable parameters. It has been well demonstrated that the approaches combining fuzzy systems with neural networks can improve the accuracy of fuzzy systems that severely depend on the experts’ knowledge to define fuzzy rules by training the parameters in an iterative way like neural networks. On the other hand, the neuro-fuzzy approaches can use the fuzzy rules learned from input–output data to increase the transparency of a neural network in producing a specific result.

The interpretability of the neural network is attracting more and more attention for its successful applications in real-world problems. However, there is no mathematical definition of interpretability until now. Interpretability can be viewed as the degree to which people can understand the cause of a decision or predict a model’s result consistently. A model is better interpretable than another one if its decisions are easier for a human to comprehend than those from the other one [2]. When we turn to the interpretability of a fuzzy system, the comprehensibility of fuzzy partition and simplicity of fuzzy rules are usually involved and discussed, which is also regarded as structural/syntactical simplicity in some papers [3], [4] to avoid the confusion.

The most popular type of fuzzy systems employed in the neuro-fuzzy models is the Takagi–Sugeno–Kang (TSK) fuzzy system [5] because of its low computational complexity and high interpretability. There are numerous models based on TSK fuzzy models and different types of shallow and deep neural networks have been proposed in recent years and applied to diverse scenarios, including time series prediction [6], [7], regression [8], [9] and classification [10]–[16], and system identification and control [17]–[20].

To achieve high accuracy, plenty of variants of neural networks are introduced into the TSK fuzzy system, therefore, the architectures of neuro-fuzzy models become more and more complex. The fuzzy wavelet polynomial neural network [21] introduces wavelet functions to the consequent part of fuzzy

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rules and combines a group of TSK fuzzy systems into one network in the same way of constructing a polynomial neural network. The fused fuzzy deep neural network [22] put a multiple-layer feed-forward neural network on top of a fuzzy system to handle the vector consisting of original input data and the output data of the fuzzy system. The hierarchical hybrid fuzzy neural network [23], [24] forms a hierarchical structure by fuzzy subsystems to aggregate some discrete attributes into one intermediate value randomly for dimension reduction. Moreover, the fuzzy restricted Boltzmann machine [16], [25], [26] and fuzzy deep belief network [27] are designed for dealing with high dimensional data based on fuzzy numbers other than TSK fuzzy systems to avoid the “rule explosion” problem. However, chasing high accuracy will inevitably increase the model complexity; consequently, the model transparency of these neuro-fuzzy models will be greatly decreased. For instance, many aforementioned approaches employ more than one fuzzy system, and fuzzy rules from different fuzzy systems are usually not compatible, sometimes they are even contradicted with each other. To reduce fuzzy rules and speed up the training process, clustering algorithms, such as k-means and fuzzy c-means, are frequently adopted to cluster the training data and derive the corresponding fuzzy sets [28]. The learning procedure of neural networks can produce suitable fuzzy rules automatically without human intervention, but it is still difficult to guarantee the proper overlap between fuzzy sets, and the input universe of discourse may not be covered by the fuzzy sets.

To improve the model simplicity of neuro-fuzzy models and retain acceptable accuracy, there are usually two factors needed to be carefully considered [29], [30]: 1) Fuzzy partition transparency, i.e., fuzzy sets should be distinguishable and normal. They should cover the universe of discourse and there is a moderate number per variable; 2) Simplicity of fuzzy rules, i.e., the number of fuzzy rules should be appropriate and they should not be too long to understand. Some models have been designed and applied to classification and function approximation. The neural fuzzy system in [30] uses a variance-based self-splitting clustering algorithm to determine the initial form of a fuzzy rule based on the distribution of input–output data, which can reduce the number of rules. The minimax probability decision technique is introduced to train a minimax probability TSK fuzzy system classifier [31] that possesses a high level of transparent model reliability. The online clustering and fuzzy set merging algorithm and multiobjective objective functions are employed in [32] to develop an efficient approach for multiobjective fuzzy control of a mobile robot. The deep TSK fuzzy classifier (D-TSK-FC) [33] and blockwise combination TSK fuzzy classifier [34] take advantage of the hierarchical structure and randomly select input features with a randomly antecedent fuzzy set selected from a fixed fuzzy partition to balance the model accuracy and complexity. Based on the same strategy, a highly interpretable D-TSK-FC (HID-TSK-FC) [35] is further developed with shared linguistic fuzzy rules. A novel fuzzy neural network with interpretable, intuitive, and correlated-contours fuzzy rules named IC-FNN is presented in [36] that may cover each hill in the function landscape to achieve a satisfactory approximation result.

Although a huge number of fuzzy and neural fuzzy systems have been developed to balance the accuracy and model complexity, it is still a big challenge to achieve a highly accurate result with satisfying fuzzy rules, or on the contrary, to possess an excellent transparent model structure with explainable rules while retaining a high accuracy, since there is always a tradeoff between them [3], [4]. However, we may expect that a compact structure could result in better transparency than models with some complex components and learning algorithms.

The fuzzy broad learning system (FBLS) [37] that retains a light structure has been proposed recently on the basis of broad learning system (BLS) [38]. The FBLS takes advantage of the ensemble of TSK fuzzy subsystems and a flat structure without any layers stacked in a deep manner to achieve a much higher accuracy in classification and regression tasks using a smaller number of fuzzy rules in total than some state-of-the-art neuro-fuzzy models. However, the fuzzy rules of an FBLS are generated from different fuzzy systems, which may not form an integrated rule base, since there are no constraints on how the fuzzy rules are produced in every fuzzy subsystem. Therefore, the model complexity of an FBLS is not satisfactory enough to match its high accuracy.

Considering the accuracy–complexity tradeoff, we will concentrate on developing a synthetically simplified FBLS in this article, named compact FBLS (CFBLS). The following strategies are adopted.

- 1) The “ensemble” structure of the FBLS is replaced with one typical TSK fuzzy system to reduce the inconsistency among fuzzy rules and model complexity.
- 2) The input universe of discourse is equally partitioned to obtain the fuzzy sets with proper linguistic labels other than determining them by clustering methods.
- 3) To reduce the total number of fuzzy rules and computational burden, we employ two matrices: The random feature selection matrix and the random rule combination matrix to determine which input feature is chosen and which fuzzy set for the selected input feature is used to generate a fuzzy rule. Further, the fuzzy rules are also randomly chosen for calculating the output of the TSK fuzzy system to avoid the “rule explosion” problem.
- 4) The enhancement nodes are kept to help bring nonlinear transformation to the first-order polynomial in the consequent part of a fuzzy rule. The parameters consisting of the weights between the enhancement layer and the top layer and the coefficients of the polynomials are also determined by pseudoinverse that guarantees the efficiency of the CFBLS.

The developed CFBLS is evaluated by some popular benchmarks from UCI and KEEL repositories, and compared with some representative fuzzy and neuro-fuzzy models, many of which are composed of deep structures and sophisticated hierarchical layers. The results reveal that the CFBLS can achieve the highest testing accuracies with a much smaller number of fuzzy rules. And the fuzzy rules generated in training process can be easily expressed and understood, which demonstrates that the CFBLS possesses excellent model simplicity without decreasing its performance.

The rest of this article is organized as follows. Section II presents the general formulation of TSK fuzzy systems and FBLS. Section III describes the proposed CFBLS. In Section IV, we make a comparison of the performance of CFBLS with other fuzzy and neuro-fuzzy models. Finally, Section V concludes this article.

II. PRELIMINARIES

A. TSK Fuzzy Systems

The TSK fuzzy system [5] is the most popular approach in fuzzy theory due to its excellent performance in numerous real-world applications. It has been widely recognized as the efficient building blocks in establishing a great deal of fuzzy and neuro-fuzzy frameworks recently. The main characteristic of a TSK fuzzy system lies in the THEN (consequent) part of a fuzzy rule that is represented by a function of input variables, which is usually a polynomial. Specifically, given an input vector $\mathbf{x} = (x_1, x_2, \dots, x_M)$, the k th fuzzy rule in a TSK fuzzy system has the following form:

$$\begin{aligned} &\text{If } x_1 \text{ is } A_{k1} \wedge x_2 \text{ is } A_{k2} \wedge \dots \wedge x_M \text{ is } A_{kM} \\ &\text{then } z_k = f_k(\mathbf{x}), k = 1, 2, \dots, K \end{aligned} \quad (1)$$

where A_{kj} is a fuzzy subset defined on the input universe of discourse, $f_k(\mathbf{x})$ is a polynomial with input variable $\mathbf{x}$, and K is the number of fuzzy rules. Generally, we set $f_k(\mathbf{x})$ be a constant or a linear combination of the components of $\mathbf{x}$, namely the zero-order or first-order TSK fuzzy system, respectively.

The output of a TSK fuzzy system can be obtained by a simple defuzzification process, that is

$$\hat{y} = \frac{\sum_{k=1}^K \prod_{j=1}^M \mu_{kj}(x_j) f_k(\mathbf{x})}{\sum_{k=1}^K \prod_{j=1}^M \mu_{kj}(x_j)} = \frac{\sum_{k=1}^K \mu_k(\mathbf{x}) z_k}{\sum_{k=1}^K \mu_k(\mathbf{x})} \quad (2)$$

where $\mu_{kj}(\cdot)$ is the membership function associated with fuzzy subset A_{kj} , and $\mu_k(\mathbf{x}) = \prod_{j=1}^M \mu_{kj}(x_j)$ represents the fire strength (activating level) of the k th fuzzy rule.

In a typical TSK fuzzy system, we need to calculate the parameters in the membership function μ_{kj} (such as the center and width of a Gaussian membership function) and the coefficients of the polynomial f_k . Traditionally, they can be determined by some popular gradient descent algorithms iteratively according to the difference between the actual target and the output of a TSK fuzzy system, however, the training process is usually very time-consuming when the input data have a large size. To speed up the training process and improve its performance, many efficient learning algorithms have been proposed [39]–[41]. For instance, the parameters of membership function can be obtained by some clustering method such as k-means and fuzzy c-means algorithms. The coefficients of the polynomial in the consequent part of a fuzzy rule can be calculated by least square method (or pseudoinverse) at once.

B. Fuzzy Broad Learning Systems

The FBLS [16] is proposed based on the BLS and an ensemble structure of TSK fuzzy subsystems. As a neuro-fuzzy structure,

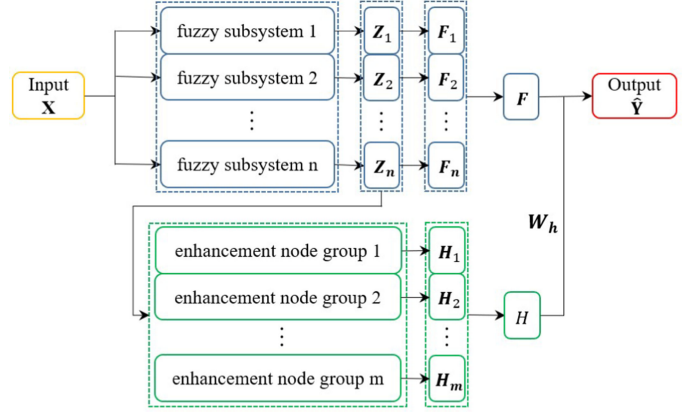


Fig. 1. Structure of an FBLS.

it inherits the fine properties of BLS, such as fast training phase and low computational complexity. Some learning algorithms for FBLS are investigated and compared in [42] and [43], where different clustering methods are employed to determine the parameters of membership functions (MFs) in the antecedent part of fuzzy rules and the coefficients in the consequent part of fuzzy rules are computed analytically by pseudoinverse. Compared with other popular fuzzy neural networks and fuzzy classifiers, the FBLS shows higher accuracy in nonlinear system identification, function regression, and classification problems. The structure of an FBLS is depicted in Fig. 1, and we will give its details in the following part.

Suppose we have n fuzzy subsystems and m sets of enhancement units in an FBLS. Given a training dataset

$$\mathbf{X} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \mid \mathbf{x}_s = (x_{s1}, x_{s2}, \dots, x_{sM}), s = 1, 2, \dots, N \right\}$$

where N and M denote the number of samples and the number of features, respectively. The corresponding target is $\mathbf{Y} = (y_1, \dots, y_N)^T \in \mathbb{R}^N$.

Suppose that K_i fuzzy rules are used in the i th first-order TSK fuzzy subsystem, which have the form of (1). We define the consequent part of k th fuzzy rule

$$z_{sk}^i = f_k^i(\mathbf{x}_s) = \sum_{t=1}^M p_{kt}^i x_{st} \triangleq \mathbf{p}_k^i \mathbf{x}_s^T \quad (3)$$

where $\mathbf{p}_k^i = (p_{k1}^i, p_{k2}^i, \dots, p_{kM}^i)$ denotes the coefficients that are randomly generated by the uniform distribution on interval $[0, 1]$.

Now the following weighted fire strength can be obtained for every fuzzy rule:

$$\bar{\mu}_{sk}^i = \frac{\prod_{t=1}^M \mu_{kt}^i(x_{st})}{\sum_{k=1}^{K_i} \prod_{t=1}^M \mu_{kt}^i(x_{st})} \quad (4)$$

where we set $\mu_{kt}^i(x_{st}) = \exp(-(\frac{x_{st} - c_{kt}^i}{\sigma_{kt}^i})^2)$.

The center c_{kt}^i and width σ_{kt}^i are determined by applying clustering approaches to the training data $\mathbf{X}$ for every fuzzy subsystem. And the number of fuzzy rules K_i of the i th fuzzy subsystem is the same as the number of clusters.

Before the defuzzification process in the TSK fuzzy subsystems, we denote $\mathbf{Z}_{si}$ the weighted outputs of fuzzy rules for the i th fuzzy subsystem and the s th training sample $\mathbf{x}_s$, that is

$$\mathbf{Z}_{si} = (\bar{\mu}_{s1}^i z_{s1}^i, \bar{\mu}_{s2}^i z_{s2}^i, \dots, \bar{\mu}_{sK_i}^i z_{sK_i}^i) \quad (5)$$

thus for all training data $\mathbf{X}$, we have

$$\mathbf{Z}_i = (\mathbf{Z}_{1i}^T, \mathbf{Z}_{2i}^T, \dots, \mathbf{Z}_{N_i}^T)^T \in \mathbb{R}^{N \times K_i}, i = 1, 2, \dots, n. \quad (6)$$

And the weighted output matrix for all the fuzzy rules and fuzzy subsystems

$$\mathbf{Z} = (\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_n) \in \mathbb{R}^{N \times \sum_{i=1}^n K_i}. \quad (7)$$

Now we begin the defuzzification process for the i th fuzzy subsystem. For a training sample $\mathbf{x}_s$, similar to (2), we have the following defuzzification result:

$$\mathbf{F}_{si} = \sum_{k=1}^{K_i} \bar{\mu}_{sk}^i \left(\sum_{t=1}^M w_{kt}^i z_{st}^i \right) = \mathbf{Z}_{si} \mathbf{w}^i \quad (8)$$

where the parameter w_{kt}^i is introduced to adjust the consequent part of the k th fuzzy rule instead of training p_{kt}^i for the purpose of reducing computational complexity, and define $\mathbf{w}^i = (w_1^i, w_2^i, \dots, w_{K_i}^i)^T$.

For training data $\mathbf{X}$, the defuzzification output of the i th fuzzy subsystem

$$\mathbf{F}_i = (\mathbf{F}_{1i}, \mathbf{F}_{2i}, \dots, \mathbf{F}_{N_i})^T = \mathbf{Z}_i \mathbf{w}^i \in \mathbb{R}^N. \quad (9)$$

Then the ensemble of n defuzzification results $\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_n$ are combined together, and we obtain

$$\mathbf{F} = \sum_{i=1}^n \mathbf{F}_i = (\mathbf{Z}_1, \dots, \mathbf{Z}_n) \begin{pmatrix} \mathbf{w}^1 \\ \vdots \\ \mathbf{w}^n \end{pmatrix} \triangleq \mathbf{Z} \mathbf{W}_f \in \mathbb{R}^N \quad (10)$$

where $\mathbf{W}_f \in \mathbb{R}^{\sum_{i=1}^n K_i}$ consists of the parameters w_{kt}^i in the consequent part.

The intermediate matrix $\mathbf{Z}$ is fed into every group of enhancement units. Suppose that we use L_j enhancement neurons in the j th set. We can thus derive the output of enhancement layer

$$\mathbf{H} = (\mathbf{H}_1, \mathbf{H}_2, \dots, \mathbf{H}_m) \in \mathbb{R}^{N \times \sum_{j=1}^m L_j} \quad (11)$$

where $\mathbf{H}_j = \phi(\mathbf{Z} \mathbf{W}_{e_j} + \mathbf{b}_j) \in \mathbb{R}^{N \times L_j}$ is the output of the j th group of enhancement units, $\mathbf{W}_{e_j}$ and $\mathbf{b}_j$ denote weights and bias terms that are uniformly generated from $[0, 1]$ ($j = 1, 2, \dots, m$) between $\mathbf{Z}$ and corresponding enhancement units, and ϕ is the activation function.

The ensemble output $\mathbf{F}$ of fuzzy subsystems and $\mathbf{H}$ are employed to obtain the output of FBLS, which can be expressed as follows:

$$\hat{\mathbf{Y}} = \mathbf{F} + \mathbf{H} \mathbf{W}_h = (\mathbf{Z}, \mathbf{H}) \begin{pmatrix} \mathbf{W}_f \\ \mathbf{W}_h \end{pmatrix} \quad (12)$$

where weight matrix $\mathbf{W}_h$ connects the enhancement nodes to the top layer.

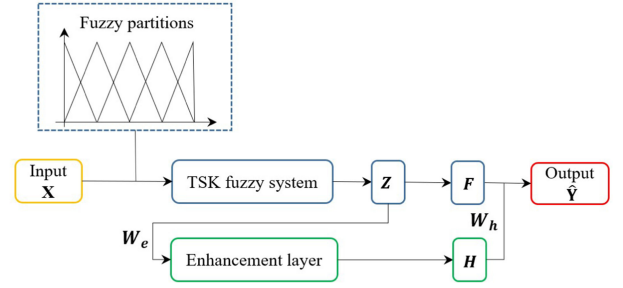


Fig. 2. Structure of a CFBLs.

Let $\mathbf{W} = (\mathbf{W}_f^T, \mathbf{W}_h^T)^T$ and $\Theta = (\mathbf{Z}, \mathbf{H})$, then it can be calculated by pseudoinverse

$$\mathbf{W} = \Theta^+ \mathbf{Y} = (\Theta^T \Theta)^{-1} \Theta^T \mathbf{Y}. \quad (13)$$

III. COMPACT FUZZY BROAD LEARNING SYSTEM

Considering the tradeoff between model complexity and accuracy, we now begin to establish the synthetically simplified version of an FBLS: The CFBLs in this section.

To achieve a high accuracy, the FBLS employs the ensemble of TSK fuzzy systems, and it does benefit from this integration, which has been verified in some popular benchmarks. The FBLS, as a neuro-fuzzy model, exhibits better performance in the aspect of accuracy; however, its model transparency is not as good as other traditional fuzzy models. One reason is that the ensemble of TSK fuzzy systems has a big increase on the total number of fuzzy rules, although there may be a few in every fuzzy subsystem. In addition, it is difficult to combine the fuzzy rules in each ensemble member together without any conflict while keeping them understandable. Therefore, the first strategy adopted to reduce model complexity of FBLS is to remove the ensemble structure of fuzzy system. Specifically, there is only one TSK fuzzy system in a CFBLs to guarantee the convenience of obtaining only one set of fuzzy rules.

In a CFBLs, which is illustrated in Fig. 2, we turn back to the classic fuzzy partition operation: The input universe of discourse is equally partitioned into G parts, where G is usually selected from $\{2, 4, 6\}$. For instance, $G = 4$ in Fig. 2 implies that for each feature there are five fuzzy subsets which have linguistic labels: Very low (MF_1), low (MF_2), medium (MF_3), high (MF_4), and very high (MF_5) from left to right. And the $G + 1$ partition points determine the shapes of five fuzzy subsets. Compared with using some clustering method to obtain fuzzy sets, this approach can provide better coverage of the entire universe of discourse and it will bring about more appropriate intersections among the resulting fuzzy sets.

For training data $\mathbf{X}$ as given in the above section, it is easy to see that there will be M^{G+1} fuzzy rules in total, which will cause the “rule explosion” problem and the increase of complexity as the number of features M grows bigger and bigger. Therefore, we introduce the second strategy: random selection of features and fuzzy rules motivated by [35] and [44], where a random rule-combination matrix and a “don’t care” matrix are used to reduce the number of fuzzy rules as well as the computational burden, and fixed centers and random widths are used for Gaussian MF.

每个特征只选一个模糊集参与规则的计算

Suppose that $\mathbf{S} \in \mathbb{R}^{M \times K}$ and $\mathbf{R} \in \mathbb{R}^{M \times (G+1) \times K}$ represent the feature selection matrix and rule combination matrix, respectively, where K is the total number of fuzzy rules. The entries of $\mathbf{S}$ and $\mathbf{R}$ are either 0 or 1, and $\sum_{g=1}^{G+1} \mathbf{R}(t, g, k) = 1$. To be clear, $\mathbf{S}(t, k) = 1$ means the t th feature is selected for the k th fuzzy rule, otherwise the t th feature will be ignored; $\mathbf{R}(t, g, k) = 1$ means that the t th feature in the k th fuzzy rule activates the g th fuzzy subsets ordered from left to right.

For instance, let $K = 2$, $M = 4$, and $G = 4$, which implied there are two fuzzy rules, four features for each sample $\mathbf{x}_s$, and five fuzzy subsets with linguistic meanings: Very low, low, medium, high, and very high. Assume that

$$\mathbf{S} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \mathbf{R}(:, :, 1) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix} \quad (14)$$

and

$$\mathbf{R}(:, :, 2) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}. \quad (15)$$

Thus we have the following two fuzzy rules.

If x_{s1} is very low, $\wedge x_{s2}$ is very high, and $\wedge x_{s4}$ is low, then $z_{s1} = \mathbf{p}_1 \mathbf{x}_s^T$.

If x_{s1} is very high $\wedge x_{s3}$ is medium, then $z_{s2} = \mathbf{p}_2 \mathbf{x}_s^T$.

Since there is only one fuzzy system (i.e., $n = 1$) in a CFBLs, we will omit the subscript or superscript i for all the symbols in the previous section from now on. The centers of Gaussian MFs for the $G + 1$ fuzzy subsets are determined by the $G + 1$ partition points, and the widths are randomly initialized within $(0, 1)$, which are define as follows:

$$\mu_g(x) = \exp \left(- \left(\frac{x - c_g}{\sigma_g} \right)^2 \right), g = 1, 2, \dots, G + 1. \quad (16)$$

Therefore, we have the following intermediate output of the TSK fuzzy system for all training data $\mathbf{X}$:

$$\begin{aligned} \mathbf{Z} &= (\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_N)^T \\ &= \begin{pmatrix} \bar{\mu}_{11} z_{11} & \bar{\mu}_{12} z_{12} & \cdots & \bar{\mu}_{1K} z_{1K} \\ \bar{\mu}_{21} z_{21} & \bar{\mu}_{22} z_{22} & \cdots & \bar{\mu}_{2K} z_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{\mu}_{N1} z_{N1} & \bar{\mu}_{N2} z_{N2} & \cdots & \bar{\mu}_{NK} z_{NK} \end{pmatrix} \end{aligned} \quad (17)$$

where $z_{sk} = \mathbf{p}_k \mathbf{x}_s^T$ and

$$\bar{\mu}_{sk} = \frac{\prod_{t=1}^M (1 - (1 - \sum_{g=1}^{G+1} \mu_g(x_{st}) \mathbf{R}(t, g, k)) \mathbf{S}(t, k))}{\sum_{k=1}^K \prod_{t=1}^M (1 - (1 - \sum_{g=1}^{G+1} \mu_g(x_{st}) \mathbf{R}(t, g, k)) \mathbf{S}(t, k))}. \quad (18)$$

Algorithm 1: Training Algorithm of the CFBLs.

Input: Training data $\mathbf{X} \in \mathbb{R}^{N \times M}$ and targets $\mathbf{Y} \in \mathbb{R}^N$, K fuzzy rules, fuzzy partition number G , and L enhancement nodes.

Output: An CFBLs with parameter matrix $\mathbf{W}$.

- 1: Initialize the coefficients $\mathbf{p}_k$ by uniform distribution in $[0, 1]$.
- 2: Randomly initialize the width σ_g within $(0, 1)$.
- 3: Randomly initialize $\mathbf{W}_e$ and $\mathbf{b}$ within $[0, 1]$.
- 4: Set center $c_g = (g - 1)/G$.
- 5: Initialize the entries of feature selection matrix $\mathbf{S}$ and rule combination matrix $\mathbf{R}$ with 0 or 1, satisfying $\sum_{g=1}^{G+1} \mathbf{R}(t, g, k) = 1$.
- 6: **for** $k = 1$ to K **do**
- 7: **for** $s = 1$ to N **do**
- 8: Calculate the weighted fire strength $\bar{\mu}_{sk}$ of the k th fuzzy rule for training sample $\mathbf{x}_s$ according to (18).
- 9: **end for**
- 10: **end for**
- 11: For all training data $\mathbf{X}$, calculate the intermediate output $\mathbf{Z}$ according to (17), the defuzzification output $\mathbf{F}$ according to (19), and the output of enhancement layer $\mathbf{H}$ according to (20);
- 12: Calculate $\mathbf{W}$ according to (13).

And the defuzzification result is

$$\mathbf{F} = (\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_N)^T = \mathbf{Z} \mathbf{w} \quad (19)$$

where $\mathbf{w} = (w_1, w_2, \dots, w_K)$ denotes the parameters to determine the coefficients $\mathbf{p}_k$ in the consequent parts ($k = 1, 2, \dots, K$).

We then also send the intermediate matrix $\mathbf{Z}$ to the enhancement layer. Suppose that there are L enhancement neurons. We can obtain the output of enhancement layer

$$\mathbf{H} = \phi(\mathbf{Z} \mathbf{W}_e + \mathbf{b}) \in \mathbb{R}^{N \times L} \quad (20)$$

where $\mathbf{W}_e$ and $\mathbf{b}$ denote weight matrix and bias terms between $\mathbf{Z}$ and enhancement layer that are uniformly generated from $[0, 1]$.

Now the output of a CFBLs can be calculated by

$$\hat{\mathbf{Y}} = \mathbf{F} + \mathbf{H} \mathbf{W}_h = (\mathbf{Z}, \mathbf{H}) \begin{pmatrix} \mathbf{w} \\ \mathbf{W}_h \end{pmatrix} \triangleq \Theta \mathbf{W} \quad (21)$$

and we can also deduce $\mathbf{W}$ according to (13).

We summarize the training algorithm of CFBLs and illustrate it in Algorithm 1.

Remark 1: After obtaining the parameters $\mathbf{W}$, the randomly initialized coefficient p_{kt} will turn into $w_k p_{kt}$, which is more appropriate for the fuzzy rules.

Remark 2: The computational complexity of generating the feature selection matrix $\mathbf{S}$ and rule combination matrix $\mathbf{R}$ is $O(MK)$ and $O(MK(G + 1))$, respectively. The computational complexity of computing $\mathbf{F}$ is $O(NMK^2(G + 1))$, and the computational complexity of computing $\mathbf{H}$ is $O(NKL)$.

$O((K + L)^3)$ is needed for calculating $\mathbf{W}$. Thus the total computational complexity of CFBLs can be derived as

$$O(NMK^2(G + 1) + NKL + (K + L)^3). \quad (22)$$

It can be easily seen that the CFBLs has much lower computational complexity than the FBLS due to the use of one TSK fuzzy system.

Remark 3: We can see from (21) that the final output $\hat{\mathbf{Y}}$ of a CFBLs has an additional term $\mathbf{H}\mathbf{W}_h$, which is different from the traditional TSK fuzzy system. However, we can easily rewrite the formula of $\hat{\mathbf{Y}}$ in accordance with the conventional fuzzy system as follows.

For simplicity, we consider only one training sample $\mathbf{x}_s$, and its corresponding output $\hat{y}_s$ can be computed by

$$\begin{aligned} \hat{y}_s &= \mathbf{F}_s + \mathbf{H}_s \mathbf{W} = \mathbf{Z}_s \mathbf{w} + \phi(\mathbf{Z}_s \mathbf{W}_e + \mathbf{b}) \mathbf{W} \\ &= \sum_{k=1}^K \bar{\mu}_{sk} (w_k \mathbf{p}_k \mathbf{x}_s^T) + \sum_{k=1}^K \frac{\phi(\mathbf{Z}_s \mathbf{W}_e + \mathbf{b}) \mathbf{W}}{K} \\ &= \sum_{k=1}^K \bar{\mu}_{sk} \left(w_k \mathbf{p}_k \mathbf{x}_s^T + \frac{\phi(\mathbf{Z}_s \mathbf{W}_e + \mathbf{b}) \mathbf{W}}{K \bar{\mu}_{sk}} \right) \\ &\triangleq \sum_{k=1}^K \bar{\mu}_{sk} \bar{z}_{sk} \end{aligned} \quad (23)$$

前件是归一化激活强度，后件是关于x的非线性函数而不是简单的多项式

where $\bar{z}_{sk} = w_k \mathbf{p}_k \mathbf{x}_s^T + \frac{\phi(\mathbf{Z}_s \mathbf{W}_e + \mathbf{b}) \mathbf{W}}{K \bar{\mu}_{sk}} \triangleq \varphi_k(\mathbf{x}_s)$, and it can be viewed as a nonlinear function of input data $\mathbf{x}_s$ used in the consequent part of the k th fuzzy rule rather than a simple polynomial. Its parameters consist of w_k , $\mathbf{W}_e$, $\mathbf{b}$, and $\mathbf{W}$, which can be determined according to Algorithm 1.

Then the k th fuzzy rule of a CFBLs can be rewritten as

$$\text{If } x_{s1} \text{ is } A_{k1} \wedge \dots \wedge x_{sM} \text{ is } A_{kM} \text{ then } \bar{z}_{sk} = \varphi_k(\mathbf{x}_s) \quad (24)$$

which is consistent with a classical TSK fuzzy model, except that the consequent part is not a polynomial now.

Or in another simple way, the additional term $\mathbf{H}\mathbf{W}_h$ can be viewed as a special “bias term” associated with the original defuzzification output $\mathbf{F}$ so as to make it more accurate, where the k th fuzzy rule still has the form

$$\text{If } x_{s1} \text{ is } A_{k1} \wedge \dots \wedge x_{sM} \text{ is } A_{kM} \text{ then } z_{sk} = \mathbf{p}_k \mathbf{x}_s^T. \quad (25)$$

IV. PERFORMANCE EVALUATION

The proposed CFBLs is evaluated on 26 datasets for classification, which are selected from [4] and [35], including both binary and multiple classification, and a real-world problem. We compare the CFBLs with some representative neuro-fuzzy models proposed recently. Specifically, we choose the following approaches:

- 1) support vector machine (SVM) [45] (linear and Gaussian kernel) as baseline;
- 2) deep belief network (DBN [46]);
- 3) fuzzy rule-based classifiers (FC-Adaboost [47], HF-GBML [48], and D-MOFRAC [4]);
- 4) hierarchical fuzzy classifier (t-HFC [49] and HID-TSK-FC [35]);

TABLE I
DETAILS OF UCI AND KEEL DATASETS FOR CLASSIFICATION

Datasets	No. of samples	No. of features	No. of classes
Diabetes	768	8	2
Australia	690	14	2
Breast	699	9	2
Adult	45222	14	2
Musk	6598	166	2
Magic04	19020	10	2
Sonar	208	60	2
Vote	232	16	2
Wpbc	198	33	2
Liver	345	6	2
Seismic	2584	15	2
Spectfheart	267	44	2
Balance	625	4	3
Pageblocks	5472	10	5
Monk2	432	6	2

- 5) TSK fuzzy system (TSK-FC [50], L2TSK-FC [50], and FS-FCSVM [51]).

The experimental results of the aforementioned models are directly cited from [4] and [35] for a fair comparison.

A. Parameter Settings for CFBLs

There is only one TSK fuzzy subsystem in CFBLs, which is different from the original FBLS. The universe of discourse of input space is uniformly partitioned into G parts, where $G \in [2, 4, 6]$. The total number of fuzzy rules K for input variables is chosen from $\{2, 3, \dots, 20\}$, while the number of enhancement nodes N_e lies in $[2, 100]$. The same training strategy is adopted here for CFBLs to determine the optimal parameters as [35] for a fair comparison, i.e., a ten-fold cross-validation with grid search during learning. All the experiments are repeated for 30 times on every dataset with different initial conditions. We report the average value, the max value, the standard deviation of classification accuracy, and the average number of fuzzy rules. The trials are implemented on a desktop computer with i7-6700 3.4GHz CPU, 16GB memory, and Windows 10 64bit OS.

B. Datasets for Benchmark

We select 15 datasets from the UCI [52] and KEEL [53] repositories. The datasets have different types of variables (features) including integer, real, and categorical numbers, and their details are displayed in Table I.

The average and maximum classification accuracies for the proposed CFBLs and other 10 models on the 15 datasets are reported in Table II. The comparison result on the average number of fuzzy rules is summarized in Table III.

We can conclude from Tables II and III the following.

- 1) The proposed CFBLs achieves the highest average accuracy on 13 of the 15 datasets and the highest maximum accuracy on nine of the 15 datasets among the models involved. SVM with linear kernel obtains the highest average accuracy on the Adult dataset, and SVM with Gaussian kernel performs the best on Pageblocks dataset. However, we can see that the CFBLs still exhibits the second best

TABLE II
PERFORMANCE COMPARISON ON CLASSIFICATION ACCURACY OF HF-GBML, TSK-FC, L2TSK-FC, FS-FCSVM, t-HFC, HID-TSK-FC, AND CFBLs ON 15 UCI AND KEEL DATASETS

Datasets	Accuracy	FC-Adaboost	DBN	SVM (Linear)	SVM (Gaussian)	HF-GBML	TSK-FC	L2TSK-FC	FS-FCSVM	t-HFC	HID-TSK-FC	CFBLs
Diabetes	Max	0.7325	0.7369	0.7135	0.7496	0.7396	0.7135	0.6425	0.6787	0.7029	0.7424	0.7890
	Aver±Std	0.7139±0.0255	0.7133±0.0147	0.6975±0.0121	0.7099±0.0274	0.7109±0.0197	0.6687±0.0327	0.6432±0.0107	0.6605±0.0168	0.6714±0.0352	0.7182±0.0156	0.7764±0.0038
Australia	Max	0.8197	0.8742	0.7842	0.7633	0.8720	0.7778	0.8293	0.8148	0.8537	0.8647	0.8870
	Aver±Std	0.8037±0.0155	0.8632±0.0112	0.7291±0.0399	0.7314±0.0243	0.8561±0.0126	0.7298±0.0399	0.7768±0.0473	0.7688±0.0501	0.8111±0.0261	0.8337±0.0187	0.8750±0.0036
Breast	Max	0.9542	0.9632	0.9602	0.9507	0.9587	0.9364	0.9587	0.9618	0.9476	0.9666	0.9771
	Aver±Std	0.9427±0.0572	0.9486±0.0136	0.9171±0.0459	0.9062±0.0371	0.9510±0.0056	0.8846±0.0432	0.9469±0.0113	0.9498±0.0142	0.9316±0.0094	0.9512±0.0081	0.9750±0.0010
Adult	Max	0.8326	0.7988	0.8395	0.8402	0.8206	0.7671	0.8015	0.7615	0.7813	0.7925	0.8301
	Aver±Std	0.8180±0.0120	0.7893±0.0088	0.8378±0.0017	0.8377±0.0016	0.8158±0.0042	0.7624±0.0028	0.7995±0.0018	0.7607±0.0005	0.7684±0.0164	0.7788±0.0064	0.8212±0.0034
Musk	Max	0.1545	0.8563	0.8474	0.8479	0.2396	0.8695	0.8574	0.8491	0.8713	0.8921	0.8885
	Aver±Std	0.1541±0.0003	0.8482±0.0086	0.8458±0.0013	0.8466±0.0013	0.1646±0.0562	0.8685±0.0013	0.8488±0.0050	0.8460±0.0019	0.8631±0.0074	0.8763±0.0153	0.8786±0.0046
Magic04	Max	0.8155	0.7926	0.6480	0.6490	0.8233	0.6498	0.7475	0.6490	0.7722	0.8241	0.8557
	Aver±Std	0.8038±0.0072	0.7889±0.0096	0.6472±0.0011	0.6473±0.0010	0.8151±0.0069	0.6487±0.0009	0.7424±0.0048	0.6480±0.0008	0.7800±0.0047	0.8160±0.0051	0.8533±0.0016
Sonar	Max	0.4807	0.7148	0.5348	0.5348	0.4808	0.5455	0.7273	0.6096	0.6812	0.7280	0.8169
	Aver±Std	0.4663±0.0083	0.6731±0.0136	0.4824±0.0413	0.5294±0.0053	0.4087±0.0797	0.4930±0.0540	0.6620±0.0401	0.5572±0.0394	0.6681±0.0181	0.6585±0.0351	0.7591±0.0162
Vote	Max	0.6238	0.9541	0.9233	0.9233	0.9633	0.8772	0.9054	0.8772	0.8745	0.9284	0.9437
	Aver±Std	0.5331±0.0879	0.9100±0.0414	0.8885±0.0374	0.9049±0.0111	0.9080±0.0364	0.8645±0.0087	0.8737±0.0274	0.8573±0.0237	0.8312±0.0412	0.9113±0.0121	0.9289±0.0088
Wpbc	Max	0.7400	0.7562	0.7753	0.7865	0.7755	0.7697	0.7584	0.7921	0.7701	0.7752	0.8184
	Aver±Std	0.7068±0.0434	0.7346±0.0216	0.7584±0.0126	0.7652±0.0134	0.7022±0.0599	0.7607±0.0094	0.7213±0.0277	0.7697±0.0138	0.7541±0.0202	0.7745±0.0029	0.8042±0.0079
Liver	Max	0.6909	0.6744	0.6742	0.6968	0.6913	0.6097	0.6935	0.6258	0.6712	0.6935	0.7568
	Aver±Std	0.6501±0.0367	0.6482±0.0334	0.6458±0.0275	0.6535±0.0281	0.6508±0.0423	0.5523±0.0573	0.6335±0.0490	0.5574±0.0502	0.6503±0.0218	0.6563±0.0212	0.7464±0.0062
Seismic	Max	0.9303	0.9412	0.9364	0.9355	0.9349	0.9347	0.9247	0.9168	0.9141	0.9333	0.9354
	Aver±Std	0.9256±0.0743	0.9309±0.0096	0.9340±0.0019	0.9343±0.0011	0.9326±0.0017	0.9320±0.0016	0.9211±0.0010	0.9023±0.0011	0.8991±0.0201	0.9321±0.0009	0.9351±0.0003
Spectfheart	Max	0.7630	0.8060	0.7958	0.8041	0.7879	0.7858	0.7817	0.7591	0.7875	0.8470	
	Aver±Std	0.7440±0.0151	0.7493±0.0641	0.7492±0.0463	0.7517±0.0432	0.7044±0.0652	0.7516±0.0346	0.7450±0.0396	0.7317±0.0396	0.7377±0.0258	0.7520±0.0344	0.8335±0.0055
Balance	Max	0.7115	0.8782	0.8843	0.9146	0.8653	0.8291	0.7580	0.7971	0.8506	0.8594	0.9119
	Aver±Std	0.6784±0.0198	0.8500±0.0173	0.8566±0.0278	0.8875±0.0294	0.8511±0.0184	0.7907±0.0416	0.7114±0.0323	0.6861±0.0803	0.8310±0.0200	0.8416±0.0155	0.9066±0.0035
Pageblocks	Max	0.8977	0.9072	0.9606	0.9606	0.9393	0.9263	0.9054	0.9157	0.9012	0.9137	0.9567
	Aver±Std	0.8976±3.1e-5	0.9003±0.0040	0.9544±0.0070	0.9551±0.0065	0.9382±0.0011	0.9025±0.0105	0.8938±0.0125	0.9082±0.0086	0.8804±0.0190	0.9067±0.0040	0.9550±0.0011
Monk2	Max	0.6667	0.6600	0.6746	0.6765	0.6623	0.6580	0.6487	0.6598	0.6478	0.6599	0.9353
	Aver±Std	0.6438±0.2061	0.6360±0.0167	0.6270±0.0459	0.6343±0.0294	0.6485±0.0125	0.6322±0.0517	0.6062±0.0535	0.6473±0.0163	0.6218±0.0186	0.6494±0.0120	0.9066±0.0149

The bold entities indicate the best result.

TABLE III
COMPARISON ON AVERAGE NUMBERS OF FUZZY RULES FOR HF-GBML, TSK-FC, L2TSK-FC, FS-FCSVM, t-HFC, HID-TSK-FC, AND CFBLs

Datasets	HF-GBML	TSK-FC	L2TSK-FC	FS-FCSVM	t-HFC	HID-TSK-FC	CFBLs
Diabetes	81.0	3.1	2.0	2.5	7.0	9.0	8.4
Australia	48.5	7.0	10.0	6.8	17.5	9.0	15.7
Breast	83.5	14.6	13.9	14.0	10.2	12.0	7.6
Adult	65.0	13.7	12.8	14.5	18.0	37.5	19.2
Musk	67.5	15.0	14.2	15.0	30.4	21.0	17.7
Magic04	86.0	13.4	10.8	11.0	28.6	36.3	18.9
Sonar	66.0	8.5	9.3	12.1	18.1	24.0	8.1
Vote	43.5	3.3	4.1	5.2	9.0	15.0	8.2
Wpbc	51.0	7.4	7.0	8.0	11.2	33.0	15.4
Liver	73.5	2.6	3.2	5.2	5.3	17.1	14.9
Seismic	26.0	7.6	8.0	8.8	5.3	25.8	6.4
Spectfheart	51.0	7.1	7.5	8.1	8.7	20.1	5.6
Balance	96.0	4.3	3.9	9.2	3.0	9.3	9.2
Pageblocks	56.0	7.2	7.0	9.3	9.0	16.2	18.9
Monk2	73.5	8.5	7.6	13.5	6.0	7.0	5.9
Average	64.5	8.2	8.1	9.6	12.5	19.5	12.0
Std.	18.0	4.1	3.7	3.6	8.1	9.8	5.1

The bold entities indicate the best result.

performance with a very slightly lower average accuracy on Pageblocks dataset.

- The CFBLs shows great improvement over other ten approaches in terms of classification accuracy. On some datasets, such as Sonar, Liver, and Monk2, the CFBLs improves the classification accuracies by nearly 10–25% compared with the recently proposed HID-TSK-FC. Moreover, the average accuracies of our model are about 5–8% higher than HID-TSK-FC on Diabetes, Spectfheart, Balance, and Pageblocks datasets.
- So far as the average number of fuzzy rules is concerned, the L2TSK-FC performs better than other models and HF-GBML has the worst performance, while the CFBLs ranks in the middle. However, the average number of fuzzy numbers of the 15 datasets for CFBLs is 12.0, which is still

much smaller than that of HID-TSK-FC. The CFBLs also has a smaller standard deviation than HID-TSK-FC, which illustrates that the CFBLs can exhibit a stable performance on the number of fuzzy rules no matter how the size of datasets and the number of sample features vary.

- Unlike HID-TSK-FC and some other recently proposed fuzzy classifiers, there is only one TSK fuzzy system in the CFBLs without any **deep or hierarchical structure**. Therefore, the fuzzy rules for classification and decision are very easy to derive from a well-trained CFBLs.

We list the seven fuzzy rules generated by the CFBLs for the Breast dataset in Table IV, and the number of fuzzy partitions $G = 4$, which implies there are five fuzzy subsets representing: Very low, low, medium, high, and very high with centers $\{0, 0.25, 0.5, 0.75, 1\}$ and widths $\{0.0409, 0.1338, 0.9349, 0.0175, 0.0611\}$. There are two classes in this dataset, then for each training sample $x_s = (x_{s1}, x_{s2}, \dots, x_{s9})$, its label y_s is denoted by $(1, -1)$ or $(-1, 1)$ in training. Therefore, the output of a CFBLs $\hat{y}_s$ is also a vector and the index of its largest entry indicates the corresponding class. In this case, the number of neurons in the enhancement layer $L = 2$. The feature selection matrix

$$S = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

TABLE IV
FUZZY RULES FOR THE BREAST DATASET

No. of rules	IF									THEN
	x_{s1}	x_{s2}	x_{s3}	x_{s4}	x_{s5}	x_{s6}	x_{s7}	x_{s8}	x_{s9}	
1	High	-	-	Low	-	-	Very high	Very low	Medium	$z_{s1} = (-6.8765, 6.8765)p_1 x_s^T$
2	-	Medium	-	-	Very high	-	-	Low	-	$z_{s2} = (-3.6867, 3.6867)p_2 x_s^T$
3	Very low	Very high	-	Medium	-	-	-	Medium	-	$z_{s3} = (-3.4994, 3.4994)p_3 x_s^T$
4	Low	-	-	Very high	-	-	Medium	Very low	Very high	$z_{s4} = (-12.4813, 12.4813)p_4 x_s^T$
5	-	Very low	Very high	-	Very low	Very high	Very low	Very high	Very low	$z_{s5} = (-6.1573, 6.1573)p_5 x_s^T$
6	-	-	-	-	-	-	Very low	-	-	$z_{s6} = (-12.1081, 12.1081)p_6 x_s^T$
7	Very high	-	Very low	Very high	Low	-	-	Low	-	$z_{s7} = (-1.4101, 1.4101)p_7 x_s^T$
p_1	0.9528	0.7041	0.9539	0.5982	0.8407	0.4428	0.8368	0.5187	0.0222	
p_2	0.3759	0.8986	0.4290	0.1996	0.3031	0.5383	0.9102	0.5253	0.3068	
p_3	0.0345	0.7153	0.7687	0.0595	0.6271	0.2652	0.3123	0.5227	0.4086	
p_4	0.8929	0.5738	0.5679	0.3945	0.6916	0.5693	0.3432	0.5949	0.2739	
p_5	0.0481	0.8381	0.1025	0.7283	0.4405	0.9972	0.5625	0.4659	0.4190	
p_6	0.2116	0.2358	0.3141	0.8699	0.2518	0.9696	0.3033	0.3740	0.3484	
p_7	0.7008	0.6144	0.7592	0.7991	0.2156	0.8936	0.8997	0.9391	0.2739	

TABLE V
COMPARISON OF CFBLs AND FBLs ON 15 UCI AND KEEL DATASETS

Datasets	CFBLs		FBLs ₁		FBLs ₂	
	Accuracy	#Rules	Accuracy	#Rules	Accuracy	#Rules
Diabetes	0.7764±0.0038	8.4	0.7868±0.0029	14.0	0.7852±0.0032	13.9
Australia	0.8750±0.0036	15.7	0.8776±0.0023	13.3	0.8756±0.0023	10.2
Breast	0.9750±0.0010	7.6	0.9756±0.0013	10.7	0.9752±0.0011	6.7
Adult	0.8212±0.0034	19.2	0.8324±0.0006	19.5	0.8151±0.0113	11.3
Musk	0.8786±0.0046	17.7	0.9096±0.0022	19.7	0.9111±0.0018	18.9
Magic04	0.8533±0.0016	18.9	0.8584±0.0004	18.9	0.8581±0.0003	18.1
Sonar	0.7591±0.0162	8.1	0.8416±0.0083	18.1	0.8481±0.0059	15.7
Vote	0.9289±0.0088	8.2	0.9621±0.0038	18.9	0.9533±0.0040	14.2
Wpbc	0.8042±0.0079	15.4	0.8124±0.0062	13.9	0.8058±0.0062	14.6
Liver	0.7464±0.0062	14.9	0.7460±0.0070	13.7	0.7445±0.0067	12.4
Seismic	0.9351±0.0003	6.4	0.9346±0.0004	10.3	0.9346±0.0002	6.3
Spectfheart	0.8335±0.0055	5.6	0.8376±0.0052	15.3	0.8361±0.0051	13.4
Balance	0.9066±0.0035	9.2	0.9171±0.0021	15.3	0.9145±0.0021	12.9
Pageblocks	0.9550±0.0011	18.9	0.9584±0.0004	19.4	0.9584±0.0004	18.8
Monk2	0.9066±0.0149	5.9	0.9718±0.0014	16.6	0.9649±0.0025	16.9
Average	-	12.0	-	15.8	-	13.6

The bold entities indicate the best result.

and for instance, the rule combination matrix for the first rule

$$R(:, :, 1) = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

The row subscripts of 1 in the first column of matrix S indicate that features x_{s1} , x_{s4} , x_{s7} , x_{s8} , and x_{s9} are selected for the first fuzzy rule, while the column subscript of 1 in each row of matrix $R(:, :, 1)$ determines the fuzzy set that every feature x_{st} fires (for instance, x_{s1} fires the fourth fuzzy set “High”).

It is worth noting that the CFBLs retains the fast computational nature of fuzzy BLS, since the parameters

are also calculated by pseudoinverse. On the contrary, other fuzzy models are usually trained in an iterative manner. Hence, the training process of CFBLs is very fast and efficient.

We now compare the performance of CFBLs with the original FBLs on the 15 datasets. Since the number of fuzzy rules is limited to 20 in the CFBLs, the total number of fuzzy rules in the FBLs is also restricted to 20 for a fair comparison. Therefore, two cases of FBLs denoted by FBLs₁ and FBLs₂ are considered: In FBLs₁, the number of fuzzy rules in each fuzzy subsystem N_r and the number of fuzzy subsystems N_s satisfy $N_r \times N_s \leq 20$ and $N_s \geq 2$, while $N_s = 1$ and $N_r \leq 20$ in FBLs₂. The testing accuracy and average number of fuzzy rules are summarized in Table V.

It can be concluded that the original FBLs can achieve much higher accuracy than CFBLs due the introduction of clustering algorithm to process the input data and the ensemble structure of fuzzy subsystems. The FBLs₁ with at least two fuzzy subsystems performs the best on 11 of 15 datasets, while FBLs₂ with only one fuzzy subsystem obtains the highest accuracy on three of them and CFBLs only beats them on two datasets.

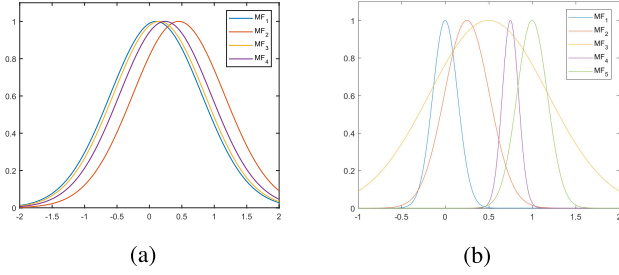


Fig. 3. Comparison of MFs on Breast dataset produced by (a) FBLs and (b) CFBLs.

It demonstrates the advantage of the ensemble structure of FBLs₁, since FBLs₂ with one fuzzy subsystem does not show a significant improvement over CFBLs. However, we can also see that FBLs₁ needs the most fuzzy rules and CFBLs has the smallest average number of fuzzy rules among them. The reduction of fuzzy rules in CFBLs can improve the model simplicity of original FBLs.

The Wilcoxon's signed-ranks test for pair-wise comparison is employed as [4] did to check if the number of fuzzy rules used in CFBLs and FBLs is statistically equivalent (null-hypothesis) with the confidence level of $\alpha = 0.05$. The p -value of applying Wilcoxon's test to CFBLs and FBLs₁ is 0.0106, while the p -value for CFBLs and FBLs₂ is 0.4296. The statistical result reveals that there is a reduction of fuzzy rules in CFBLs comparing with FBLs₁.

Furthermore, as we have mentioned before, there is always a tradeoff between model simplicity and accuracy. The fuzzy rules generated from different fuzzy subsystems by k-means algorithm in the FBLs are usually not compatible, sometimes they are even contradicted with each other, since there is no restriction in clustering the input data in order to achieve a more accurate result. And the MFs that are determined by the centers of clusters may have improper intersection and may not appropriately cover the input universe of discourse in FBLs. Therefore, we could not assign them the proper linguistic labels as we did in CFBLs. One example is shown in Fig. 3. We can see that the MFs in FBLs₁ [see Fig. 3(a)] are very concentrated and the centers do not spread out on the discourse of universe. On the contrary, the CFBLs can produce MFs with better spread centers and intersections (see Fig. 3), which can help to label them with linguistic terms: Very low, low, medium, high, and very high, and generate more understandable rules.

We also compare our CFBLs with the D-MOFRAC proposed in [4] that is also a fuzzy rule-based classifier aimed to balance the model accuracy and complexity. The following 15 datasets with multiple classes listed in Table VI are added to make a more comprehensive comparison. The results are summarized in Table VII.

It is easy to see from Table VII that our CFBLs shows higher accuracy on 17 of 22 datasets than D-MOFRAC. The smaller standard deviation of the accuracy in CFBLs indicates a much stabler performance.

Furthermore, the Wilcoxon's signed-ranks test has been used to see whether the performance of CFBLs and D-MOFRAC are statistically equivalent (null-hypothesis), considering

TABLE VI
DETAILS OF KEEL DATASETS WITH MULTIPLE CLASSES

Datasets	No. of samples	No. of features	No. of classes
Iris	150	4	3
Tae	151	5	3
Wine	178	13	3
Automobile	159	25	6
Glass	214	9	7
Newthyoid	215	5	3
Cleveland	297	13	5
Harberman	306	3	3
Ecoli	336	7	8
Vehicle	846	18	4
Contraceptive	1473	9	3
Segment	2310	19	7
Texture	5500	40	11
Satimage	6435	36	7
Penbased	10992	16	10

TABLE VII
CLASSIFICATION PERFORMANCE COMPARISON OF CFBLs AND D-MOFRAC ON UCI AND KEEL DATASETS

Datasets	CFBLs		D-MOFRAC	
	Accuracy	#Rules	Accuracy	#Rules
Diabetes	0.7764±0.0038	8.4	0.7550±0.0447	10.4
Australia	0.8750±0.0036	15.7	0.8600±0.0440	19.8
Breast	0.9750±0.0010	7.6	0.9680±0.0013	9.0
Magic04	0.8533±0.0016	18.9	0.8540±0.0550	32.2
Liver	0.7464±0.0062	14.9	0.7010±0.1051	7.7
Balance	0.9066±0.0035	9.2	0.8560±0.0326	19.8
Pageblocks	0.9550±0.0011	18.9	0.9700±0.0055	21.5
Iris	0.9822±0.0038	5.0	0.9600±0.0419	5.6
Tae	0.6406±0.0092	16.3	0.5930±0.1179	20.2
Wine	0.9740±0.0114	13.7	0.9580±0.0388	8.6
Automobile	0.7421±0.0167	20.0	0.8100±0.0856	38.9
Glass	0.7086±0.0072	16.0	0.7060±0.1324	27.4
Newthyoid	0.9508±0.0071	10.0	0.9550±0.0455	9.5
Cleveland	0.6195±0.0003	12.7	0.5290±0.0665	45.6
Harberman	0.7354±0.0669	2.0	0.6940±0.0506	9.2
Ecoli	0.8530±0.0045	17.7	0.8280±0.0484	26.2
Vehicle	0.7797±0.0112	20.0	0.7060±0.0468	22.4
Contraceptive	0.5603±0.0004	22.7	0.5360±0.0307	49.0
Segment	0.9395±0.0015	28.3	0.9660±0.0089	26.2
Texture	0.9856±0.0055	57.0	0.9520±0.0105	60.5
Satimage	0.8795±0.0063	56.7	0.8750±0.0079	56.0
Penbased	0.9828±0.0003	98.7	0.9620±0.0056	119.2
Average	-	22.3	-	28.4

The bold entities indicate the best result.

a confidence level of $\alpha = 0.05$. The p -values of applying Wilcoxon's test to the average test accuracy and the number of fuzzy rules are 0.0078 and 0.0035, respectively, which are much smaller than 0.05; thus the null-hypothesis is rejected. The statistical results also confirm that CFBLs outperforms D-MOFRAC with higher accuracy and fewer number of rules.

C. Real-World Application

The Electricity Pricing dataset [54] collects 45 312 real-world data from May 1996 to December 1998 with regular 30-min intervals to predict whether the electricity price will go up or down, which falls into a binary classification problem. Every sample has nine features and one label (1 or -1). We also use five features including the date of week, half-hour period, the demand in New South Wales, the demand in Victoria, and the

TABLE VIII
PERFORMANCE COMPARISON ON THE ELECTRICITY PRICE DATASET

Models	Accuracy		Average No. of rules
	Max	Aver $\pm$ Std	
FC-Adaboost	0.7186	0.7152 $\pm$ 0.0042	N/A
DBN	0.7202	0.6999 $\pm$ 0.0143	N/A
SVM (Linear)	0.7390	0.7362 $\pm$ 0.0024	N/A
SVM (Gaussian)	0.7398	0.7378$\pm$0.0025	N/A
FS-FCSVM	0.7113	0.6987 $\pm$ 0.0021	14.4
TSK-FC	0.7281	0.7169 $\pm$ 0.0142	13.6
HF-GBML	0.7337	0.7213 $\pm$ 0.0038	81.0
t-HFC	0.7361	0.7119 $\pm$ 0.0193	10.2
L2TSK-FC	0.7401	0.7137 $\pm$ 0.0371	15.0
HID-TSK-FC	0.7481	0.7223 $\pm$ 0.0428	21.5
CFBLS	0.7572	0.7315 $\pm$ 0.0085	17.3

The bold entities indicate the best result.

TABLE IX
AVERAGE RANKINGS OF CFBLS, DBN, SVM (LINEAR), SVM (GAUSSIAN), AND FC-ADABOOST

	FC-Adaboost	DBN	SVM(Linear)	SVM(Gaussian)	CFBLS
Ranking	3.9375	3.25	3.75	2.75	1.3125
<i>p</i> -value	1.11×10^{-5}				

amount of electricity scheduled for transfer between the two states. Twenty seven thousand five hundred and fifty two data without missing features are selected for testing our model.

The performance comparison is given in Table VIII. We can observe that the CFBLS can achieve the best maximum accuracy among all the approaches, and the SVMs have the highest average accuracy. However, the CFBLS outperforms other fuzzy models with a moderate number of fuzzy rules, which demonstrates its superiority. Especially, it needs fewer fuzzy rules and performs better in terms of maximum and average accuracies than the recently proposed HID-TSK-FC.

D. Statistical Analysis

The Friedman test and Holm post-hoc test [55] are employed to analyze the statistically significant differences among the aforementioned 11 algorithms over the 16 datasets.

First, we apply the Friedman ranking test with confidence level $\alpha = 0.05$ on the whole set of models to check whether the measured average ranks are significantly different, and the *p*-value is $1.07 \times 10^{-9} \ll 0.05$, which implies that the ranks of these models are significantly different. In addition, the one-way repeated ANOVA measure is used to verify if the average accuracies of each model differs significantly from the aggregate average value across all conditions, and the *F* score for the ANOVA test is 3.46 with *p*-value $4.13 \times 10^{-4} \ll 0.05$, which further show that the 11 models have significant different performance on different datasets.

Second, the 11 models are divided into two groups according to whether the fuzzy rules are available in the comparison: 1) DBN, SVM (linear), SVM (Gaussian), and FC-Adaboost and 2) HF-GBML, TSK-FC, L2TSK-FC, FS-FCSVM, t-HFC, and HID-TSK-FC. The Friedman test is applied to each group and the results are displayed in Tables IX and X. We can see that the

TABLE X
AVERAGE RANKINGS OF HF-GBML, TSK-FC, L2TSK-FC, FS-FCSVM, T-HFC, HID-TSK-FC, AND CFBLS

	HF-GBML	TSK-FC	L2TSK-FC	FS-FCSVM	t-HFC	HID-TSK-FC	CFBLS
Ranking	3.9375	5.125	4.9375	5.5	4.9375	2.5625	1.0
<i>p</i> -value	3.35×10^{-10}						

TABLE XI
HOLM TEST FOR CFBLS VERSUS DBN, SVM (LINEAR), SVM (GAUSSIAN), AND FC-ADABOOST WITH $\alpha = 0.05$

<i>i</i>	Models	<i>z</i>	<i>p</i>	Holm	Hypothesis
4	FC-Adaboost	4.6957	2.66×10^{-6}	0.0125	Rejected
3	SVM(Linear)	4.3603	1.30×10^{-5}	0.0167	Rejected
2	DBN	3.4659	5.28×10^{-4}	0.0250	Rejected
1	SVM(Gaussian)	2.5715	0.0101	0.0500	Rejected

TABLE XII
HOLM TEST FOR CFBLS VERSUS HF-GBML, TSK-FC, L2TSK-FC, FS-FCSVM, T-HFC, AND HID-TSK-FC WITH $\alpha = 0.05$

<i>i</i>	Models	<i>z</i>	<i>p</i>	Holm	Hypothesis
6	FS-FCSVM	5.8919	3.82×10^{-9}	0.0083	Rejected
5	TSK-FC	5.4009	6.63×10^{-8}	0.0100	Rejected
4	L2TSK-FC	5.1554	2.53×10^{-7}	0.0125	Rejected
3	t-HFC	5.1554	2.53×10^{-7}	0.0167	Rejected
2	HF-GBML	3.8461	1.20×10^{-4}	0.0250	Rejected
1	HID-TSK-FC	2.0458	0.04078	0.0500	Rejected

CFBLS has the top rank in each group and the *p*-values indicate that there are significant differences among different models.

Then the Holm post-hoc test is employed to compare CFBLS with other models in each group. The Holm's procedure rejects the hypothesis that the pair of models has no performance difference when the unadjusted *p*-value is smaller than the *Holm*. The results listed in Tables XI and XII illustrate that the CFBLS has the best performance in both groups, which demonstrates its efficiency and advantage.

V. CONCLUSION

High accuracy and excellent model simplicity are two important research topics for fuzzy systems and neuro-fuzzy models, however, they usually cannot be achieved simultaneously in one structure. On one hand, there are numerous hybrid models combining fuzzy systems with neural networks developed very recently with higher and higher accuracy, but they are also very difficult to understand due to the complex structures and abundant fuzzy rules. On the other hand, seeking models with lower complexity always results in uncompetitive performance in accuracy.

The FBLs is a recently proposed neuro-fuzzy model that shows high accuracy in both classification and regression tasks and inherits the fast computational nature of a BLS. However, the ensemble structure of fuzzy subsystems decrease its transparency since the fuzzy rules from different fuzzy systems usually cannot be combined together while keeping the consistency. In this article, we develop the CFBLS on the basis of

FBLS. The CFBLs inherits the architecture of FBLS but is more compact: the input universe of discourse is equally partitioned to obtain fuzzy sets with proper linguistic labels accordingly. The random feature selection and rule combination matrices are used to reduce the total number of fuzzy rules and avoid the “curse of dimensionality.” The enhancement layer is kept in the CFBLs that helps to achieve a competitive accuracy. The pseudoinverse is also used to determine the parameters of CFBLs, which guarantees a fast computational nature.

The experimental results on UCI and KEEL datasets, as well as a real-world case, indicate that the CFBLs can achieve much higher accuracy with a smaller number of fuzzy rules than some state-of-the-art neuro-fuzzy models, especially the deep model HID-TSK-FC, which demonstrates the superiority of CFBLs. The fuzzy rules produced by the CFBLs can be easily obtained and expressed after training.

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